

HANBURY RIVER, NT, Canada

WMO: 719630

Lat: 63.600N

Lon: 105.130W

Elev: 317

StdP: 97.57

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -41.6	(c) -39.9	(d) -45.6	(e) 0.0	(f) -41.4	(g) -43.9	(h) 0.1	(i) -40.0	(j) 16.1	(k) -22.5	(l) 14.1	(m) -26.5	(n) 2.6	(o) 270	(p) 0.970

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	23.8	15.2	21.8	13.9	19.7	13.1	16.4	21.7	15.1	19.8	13.9	18.4	5.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.4	10.6	17.8	13.0	9.7	16.8	11.8	8.9	15.6	47.2	21.6	43.3	19.9	40.1	18.4	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	12.7	11.0	-43.3	27.2	2.0	1.7	-44.7	28.4	-45.8	29.4	-47.0	30.4	-48.4	31.6
			-43.9	17.7	1.6	3.5	-45.1	20.2	-46.0	22.3	-46.9	24.2	-48.1	26.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.3	-29.1	-28.3	-23.7	-15.7	-4.2	8.3	13.4	11.1	4.1	-6.1	-18.2	-25.1
	DBStd	16.40	7.11	6.96	7.51	6.99	6.34	4.68	3.55	3.76	4.66	5.51	6.81	7.06
	HDD10.0	7281	1213	1071	1045	769	439	87	9	31	184	498	846	1089
	HDD18.3	10106	1471	1304	1303	1019	697	301	157	225	428	756	1096	1347
	CDD10.0	223	0	0	0	0	0	36	114	66	7	0	0	0
	CDD18.3	6	0	0	0	0	0	1	4	1	0	0	0	0
	CDH23.3	73	0	0	0	0	0	12	44	17	0	0	0	0
	CDH26.7	5	0	0	0	0	0	1	3	1	0	0	0	0
(14) Wind	WSAvg	4.8	4.6	4.5	4.5	4.4	4.6	4.6	5.0	4.9	5.7	5.2	5.0	4.5
(15) Precipitation	PrecAvg	292	10	9	12	11	19	31	46	56	43	28	16	11
	PrecMax	415	18	18	26	29	53	84	106	94	72	47	33	26
	PrecMin	212	3	2	3	2	3	6	21	24	12	13	7	4
	PrecStd	41	4	4	6	5	11	15	17	19	15	7	7	5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.7	-8.0	-1.8	2.3	13.6	25.1	26.9	25.5	18.2	7.9	-0.8	-5.9
		MCWB	-7.9	-8.1	-2.6	1.0	7.5	16.2	16.6	15.4	12.4	6.4	-1.0	-6.2
	2%	DB	-11.3	-11.4	-6.5	0.5	9.0	22.1	24.5	22.5	15.5	4.6	-3.4	-8.3
		MCWB	-11.5	-11.7	-7.0	-0.2	5.6	13.1	15.8	14.7	10.9	3.8	-3.6	-8.6
	5%	DB	-15.0	-14.4	-9.4	-2.2	5.7	19.5	22.6	20.0	13.1	2.4	-5.8	-11.2
		MCWB	-15.3	-14.7	-9.8	-2.9	3.6	12.7	14.2	13.4	9.5	1.7	-6.0	-11.5
	10%	DB	-18.5	-17.2	-12.2	-4.6	3.5	17.1	20.7	17.7	10.8	0.7	-8.0	-14.3
		MCWB	-18.9	-17.6	-12.6	-5.2	1.9	11.0	13.6	12.4	8.4	0.3	-8.2	-14.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.3	-8.3	-5.0	1.1	8.2	19.3	18.3	17.2	13.6	6.9	-1.2	-6.1
		MCDB	-8.2	-8.2	-4.6	2.0	12.0	20.4	23.6	22.0	16.8	8.4	-1.0	-5.9
	2%	WB	-12.2	-12.6	-7.9	-0.4	5.5	15.2	16.6	15.6	11.5	3.9	-4.2	-8.4
		MCDB	-12.0	-12.3	-7.4	0.3	8.5	19.9	22.6	20.3	14.7	4.6	-4.0	-8.2
	5%	WB	-15.7	-15.8	-10.4	-3.1	3.8	13.2	15.5	14.3	9.9	1.8	-6.8	-11.7
		MCDB	-15.4	-15.5	-10.0	-2.6	5.7	18.7	20.6	18.7	12.2	2.3	-6.6	-11.4
	10%	WB	-19.1	-18.5	-13.2	-5.4	2.0	11.6	14.4	13.2	8.5	0.4	-9.2	-14.8
		MCDB	-18.9	-18.2	-12.7	-4.8	3.3	16.6	19.0	16.9	10.5	0.8	-9.0	-14.5
(35) Mean Daily Temperature Range	MDBR		7.5	7.9	8.9	9.9	7.9	10.3	9.8	8.8	6.9	5.7	7.3	7.8
	5% DB	MCDBR	9.9	10.6	10.0	10.4	10.1	14.8	12.8	12.6	11.0	6.6	8.3	10.8
		MCWBR	9.9	11.1	9.8	9.4	6.8	8.3	5.8	6.0	6.8	5.8	9.4	11.2
	5% WB	MCDBR	10.0	10.9	10.0	9.9	9.7	13.9	11.5	11.1	10.0	6.5	9.0	11.1
		MCWBR	10.0	10.8	9.8	9.2	6.7	9.3	6.4	6.1	6.8	5.8	9.0	11.0
(40) Clear-Sky Solar Irradiance	taub		0.161	0.234	0.247	0.292	0.309	0.336	0.353	0.328	0.296	0.255	0.173	0.143
	taud		1.990	2.169	2.268	2.175	2.293	2.445	2.417	2.468	2.544	2.333	1.964	1.933
	Ebn at Noon		509	718	870	887	904	884	855	849	816	688	445	287
	Edh at Noon		27	59	81	113	111	99	99	86	65	50	27	12
(44) All-Sky Solar Radiation	RadAvg		0.19	0.94	2.71	4.95	6.11	5.80	5.26	3.87	2.26	0.73	0.25	0.08
	RadStd		0.01	0.04	0.18	0.30	0.35	0.41	0.34	0.29	0.28	0.17	0.04	0.01

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page